Lab Activity: Building BMI Calculator App using Vue CDN

Learning Outcomes

By the end of this lab, we will be able to:

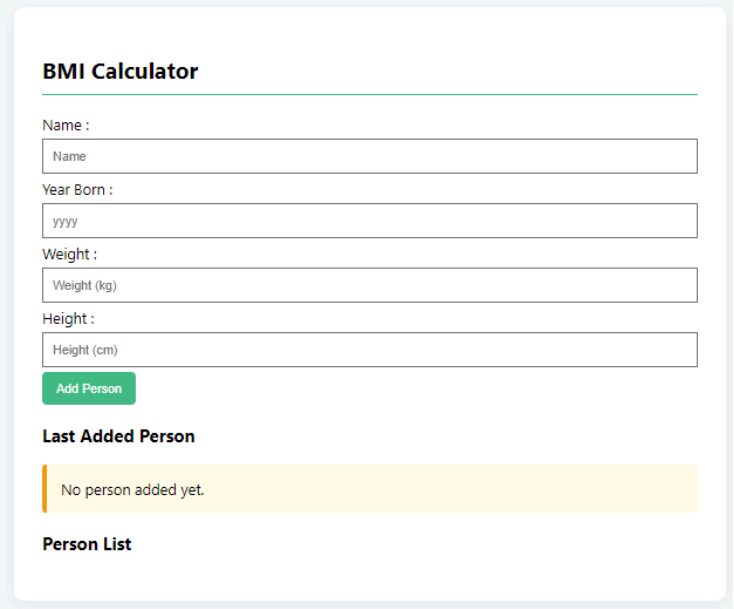
- ✔ Set up a Vue 3 app using CDN
- ✔ Understand and use core Vue concepts: vue createApp, mount, data(), methods, computed, v-model, v-on, v-if, v-for
- ✔ Handle form input and dynamic lists
- ✔ Compute age, BMI and categorize it dynamically
- ✔ (Optional) Simulate data submission to a mock backend

Activity Breakdown

Phase	Description
Phase 1 – Understand the HTML+js code	Understand current app using plain JS
Phase 2 – Vue Init & Input Binding, interpolation	Vue CDN, createApp, mount, data(){}, v-model, {{ }}
Phase 3 – Form + List Rendering	Handle submit, update array, v-for rendering
Phase 4 – BMI, age Logic & Conditional	BMI method, category class, dynamic UI
Phase 5 – Delete & Clear All	List management with delete buttons
Phase 6 – Connect to backend	API call using fetch, async/await (optional)

Step-by-Step Lab Activity

Phase 1 – Understand the Plain JavaScript App



Open the provided file bmi-vanilla.html.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BMI App (With Age)</title>
  <style>
    body {
      font-family: 'Segoe UI', sans-serif;
      background-color: #f4f7f9;
      padding: 20px;
    }
    .container {
      max-width: 700px;
      margin: auto;
      background: white;
      padding: 30px;
      border-radius: 10px;
      box-shadow: 0 4px 20px rgba(0,0,0,0.05);
    }
    h2 {
      border-bottom: 2px solid #42b983;
      padding-bottom: 10px;
    }
    form {
      margin-bottom: 20px;
    }
    input {
      padding: 10px;
      margin: 5px 0;
      width: 100%;
      box-sizing: border-box;
    }
    button {
      background-color: #42b983;
      color: white;
      padding: 10px 15px;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }
    ul {
      padding: 0;
      list-style: none;
    }
    li {
      background: #f0f9f5;
      padding: 15px;
      margin-bottom: 10px;
      border-left: 5px solid #42b983;
      border-radius: 5px;
    }
    .single {
      background: #fffbe6;
      padding: 15px;
      border-left: 5px solid #f39c12;
      border-radius: 5px;
      margin-bottom: 20px;
    }
  </style>
</head>
<body>

  <div class="container">
    <h2>BMI Calculator</h2>

    <form id="bmiForm">
      <input type="text" id="name" placeholder="Name" required />
      <input type="number" id="yob" placeholder="Year of Birth" required />
      <input type="number" id="weight" placeholder="Weight (kg)" required />
      <input type="number" id="height" placeholder="Height (cm)" required />
      <button type="submit">Add Person</button>
    </form>
```

```
    <h3>Last Added Person</h3>
    <div id="singlePerson" class="single">No person added yet.</div>

    <h3>Person List</h3>
    <ul id="personList"></ul>
  </div>

  <script>
    const personList = [];
    document.getElementById('bmiForm').addEventListener('submit', function(event) {
      event.preventDefault();
      const name = document.getElementById('name').value;
      const yob = parseInt(document.getElementById('yob').value);
      const weight = parseFloat(document.getElementById('weight').value);
      const height = parseFloat(document.getElementById('height').value);
      const bmi = (weight / ((height / 100) * (height / 100))).toFixed(2);
      const currentYear = new Date().getFullYear();
      const age = currentYear - yob;

      const person = {
        name: name,
        yob: yob,
        age: age,
        weight: weight,
        height: height,
        bmi: bmi
      };

      personList.push(person);
      displaySinglePerson(person);
      displayPersonList();
      document.getElementById('bmiForm').reset();
    });

    function displaySinglePerson(p) {
      const div = document.getElementById('singlePerson');
      div.innerHTML = `Name: ${p.name}, Age: ${p.age}, Weight: ${p.weight} kg, Height:
${p.height} cm, BMI: ${p.bmi}`;
    }

    function displayPersonList() {
      const ul = document.getElementById('personList');
      ul.innerHTML = '';
      for (let i = 0; i < personList.length; i++) {
        const p = personList[i];
        const li = document.createElement('li');
        li.textContent = `Name: ${p.name}, Age: ${p.age}, Weight: ${p.weight} kg, Height:
${p.height} cm, BMI: ${p.bmi}`;
        ul.appendChild(li);
      }
    }
  </script>

</body>
</html>
```

Phase 1 – Review the Vanilla Version
Observe inputs: name, year of birth, weight, height

Understand how:
- ✔ Age is calculated from current year and YOB
- ✔ BMI is calculated
- ✔ Last added person, ListPerson - shown separately

Identify limitations:
- ✔ Manual DOM manipulation
- ✔ DOM is tightly coupled with JS logic

✔ Hard to scale

Phase 2 – Set Up Vue
Vue Concept Quick Guide

Concept	Explanation
Vue CDN	Loads Vue directly into the browser via <script>, no build tool needed.
Vue.createApp()	Creates a new Vue app instance with data and methods.
data()	Returns reactive data (i.e., Vue watches for changes).
mount('#id')	Attaches the Vue app to an element in the HTML.
v-model	Two-way binding between input and data.
v-on / @	Event binding like @click, @submit.
v-for	Loops through an array to render lists.
v-if	Conditionally show/hide elements.
:class	Bind class name based on data logic.

Add Vue CDN script in <head>:

```
<head>
  <meta charset="UTF-8">
  <title>BMI App with Vue CDN</title>
  <script src="https://unpkg.com/vue@3/dist/vue.global.js"></script>
```

Wrap content in:

```
<div id="app">
  <!-- content here -->
</div>
```

Initialize Vue and mount to html section with id #app:

```
  const app = Vue.createApp({
  data() {
    return {
    //complete the list of data here
    }
  }
});

app.mount('#app');
```

Bind inputs using v-model:

```
<form @submit.prevent="addPerson">
      <!-- lengkapkan kod -->
      <button type="submit">Add Person</button>
    </form>
```

Using interpolation {{ }} to show live preview:

```
<p>You typed: {{ name }}, {{ yob}} ,{{ weight }} kg, {{ height }} cm</p>
```

Phase 3 – Handle Submit & Show List
Create addPerson() method:

```
addPerson() {
          const age = Date().getFullYear() - this.yob;
          const bmi = //complete this
          //lengkapkan kod here
```

Show last person added (conditional rendering using v-if):

```
<h3>Last Added Person</h3>
    <div v-if="lastPerson">
      Name: {{ lastPerson.name }},
      //etc
      //etc
    </div>
    <div class="single" v-else>No person added yet.</div>
```

Show the list using v-for:

```
h3>Person List</h3>
<ul>
  <li v-for="i in people.length" :key="i">
    Name: {{ people[i - 1].name }},
    //etc
    //etc
  </li>
</ul>
```

Phase 4 – BMI, age Logic & Conditional

Add methods:

```
calculateBMI() {
  return ????
},
calcAge(){
//return ????
}
```

Phase Last – Optional
Mock API POST:

```
await fetch('https://jsonplaceholder.typicode.com/posts', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify(person)
});
```

Loading/Error:

```
loading: false,
error: '',
```

Use v-if="loading" or v-if="error" in HTML

```
mounted() {
  const saved = localStorage.getItem('bmiList');
```

```
  if (saved) this.people = JSON.parse(saved);
}
```

code update:

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <script src="https://unpkg.com/vue@3/dist/vue.global.js"></script>
    <style>
        body {
            font-family: 'Segoe UI', sans-serif;
            background-color: #f4f7f9;
            padding: 20px;
        }

        .container {
            max-width: 700px;
            margin: auto;
            background: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 4px 20px rgba(0, 0, 0, 0.05);
        }

        h2 {
            border-bottom: 2px solid #42b983;
            padding-bottom: 10px;
        }

        form {
            margin-bottom: 20px;
        }

        input {
            padding: 10px;
            margin: 5px 0;
            width: 100%;
            box-sizing: border-box;
        }

        button {
            background-color: #42b983;
            color: white;
            padding: 10px 15px;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        ul {
            padding: 0;
            list-style: none;
        }

        li {
            background: #f0f9f5;
            padding: 15px;
            margin-bottom: 10px;
            border-left: 5px solid #42b983;
            border-radius: 5px;
        }

        .single {
            background: #fffbe6;
            padding: 15px;
            border-left: 5px solid #f39c12;
            border-radius: 5px;
```

```
            margin-bottom: 20px;
        }
    </style>
</head>

<body>

    <div id="app" class="container">
        <h2>BMI Calculator</h2>


        <form @submit.prevent="addPerson()">
            <input type="text" v-model="name" placeholder="Name" required />
            <input type="number" v-model="yob" placeholder="yyyy" required />
            <input type="number" v-model="weight" placeholder="Weight (kg)" required />
            <input type="number" v-model="height" placeholder="Height (cm)" required />
            <button type="submit">Add Person</button>
        </form>

        <h3>Last Added Person</h3>
        <!-- conditional rendering v-if & v-else -->
        <p v-if="p!=null">You typed: {{ p.name }}, {{ p.yob}} , {{ p.age }},{{ p.weight }} kg,
{{ p.height }},  {{ p.bmi}} cm</p>
        <p v-else> p is empty</p>


        <!-- <div id="singlePerson" class="single">No person added yet.</div> -->


        <h3>Person List</h3>

        <ul>
            <li v-for="i in personList.length" :key="i">
                {{ personList[i-1].name }}
              </li>
        </ul>
    </div>


</body>
<script>
    const app = Vue.createApp({
        data() {
            return {
                name: '',
                yob: 0,
                weight: 0,
                height: 0,
                category: '',
                p:null,
                personList:[]
            }
        },
        computed : {
            bmi(){
                return this.weight / ((this.height/100) ^ 2);
            },
            age(){
                return 2025-this.yob;
            }
        },
        methods: {
            addPerson() {
                // alert('savePerson clicked...');
                //create object person, mybe send/post to backend
                const p = {
                    name: this.name,
                    yob: this.yob,
                    age: this.age, //panggil computed age
                    weight: this.weight,
                    height: this.height,
                    bmi: this.bmi, //panggil method calcBmi()
                    category: 'normal'
                }
```

```
                this.p = p;
                this.personList.push(p);  //masukkan p kedalam senarai

            },
            calcAge() {
                return 2025 - this.yob;
            },
            calcBmi(){
                return this.weight / (this.height ^ 2);
            }
        }

    });
    app.mount("#app");
</script>

</html>
```